

RMIT

MIET2134 – Engineering Dynamics

**Moment of Inertia of an
Automobile Crankshaft**

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Abstract

The purpose of this experiment is, utilising experimental and analytical methods, to determine the mass moment of inertia of a crankshaft. Both a steel cylinder and a steel crankshaft are suspended from their centre axis from a single wire with their geometric data measured and trials recording the time it takes to complete a set number of oscillations. This data was used to the experimental moment of inertia to be $0.03581 \text{ kg} \cdot \text{m}^2$ and the analytical moment of inertia to be $0.1874 \text{ kg} \cdot \text{m}^2$. The experiment demonstrated how the use of two theories, the torsional pendulum and the parallel axis, can give vastly different results for the same property of an object.

Introduction

Purpose and Objectives

The purpose of this experiment was to determine that mass moment of inertia of a crankshaft using two different methods. First find the experimental MOI through its known mass and second, calculating the MOI is using the parallel-axis theorem with the assumption of the crankshaft's geometry. The mass moment of inertia as stated by Dr. Firas Jarrar is “an objects [ability] to resist change in its rotational speed about a specific axis” (Jarrar 2020, p. 2).

Background and Theory

The mass moment of inertia (MOI) outlines a mass' resistance along an axis to angular acceleration. The MOI for any solid shape can be calculated using the equation $I = \int r^2 dm$ (1), and for any mass with a sectioned body using the equation $I = \sum_i n_i (I_{cm,i} + m_i z_i^2)$ (2), where I is the moment of inertia, r is the radius, n_i is the amount of each geometric sections there is, m_i is the mass, z_i is the offset from the central axis and $I_{cm,i}$ is the moment of inertia of a section around its own centre of mass.

For a general shape such as the cylinder used the experiment, there is a more general equation to use, $I = \frac{1}{2} mr^2$ (3), which can be seen in the *Experimental MOI* section as the equation,

$I_{cylinder} = \frac{1}{2} \rho \pi r_c^4 h_c$ (4). Where ρ is the density of the material and h_c is the height of the cylinder. This is the analytical MOI of the cylinder, however the result from this equation can be used to find the stiffness of the wire represented as $k_{cylinder}$, using the equation, $f_{cylinder} =$

$\frac{1}{2\pi} \sqrt{\frac{k_{cylinder}}{I_{cylinder}}}$ (5), where $f_{cylinder}$ is the natural frequency of the cylinder and $I_{cylinder}$ is the cylinder's

moment of inertia. Utilised in the report as the equation, $k_{cylinder} = I_{cylinder} \left(\frac{2\pi}{T_{cylinder}} \right)^2$ (6),

where $T_{cylinder}$ is the period of the cylinder. Finally, from this the experimental MOI of the

crankshaft, $I_{crankshaft}$, could be estimated by, $I_{crankshaft} = \frac{k_{cylinder}}{(2\pi \times f_{crankshaft})^2}$ (7), which is found in

the report as, $I_{estimate} = k_{cylinder} \times \frac{T_p^2}{4\pi^2}$ (8), where $I_{estimate}$ is the crankshaft's moment of inertia found experimentally and T_p is the period of the crankshaft. This uses the theory of a torsional pendulum where the wire acts as a torsional spring where the torsional stiffness of the wire is used to determine the moment of inertia for the attached object.

An alternative way to approximate the moment of inertia is to analyse the geometric sections of the crankshaft. This can be done by using equation (4) and altering the variables to apply to a particular section, this is given in the report as, $m_i = \rho \pi r_i^2 h_i$ (9) and $I_{cmi} = \frac{1}{2} m_i r_i^2$ (10). I_{cmi} is the moment of inertia of a section around its centre of mass. However, this may need to be altered to lie on the axis of rotation, where we apply the parallel-axis theorem, $I_{single,i} = I_{cmi} + m_i z_i^2$ (11), where $I_{single,i}$ is the moment of inertia of a section around the axis and z_i is the offset of the section from the axis. Doing this for every section of the crankshaft gives equation (1), the total moment of inertia of the crankshaft.

These results are likely to differ as they are derived using geometric simplifications and assumptions which all impact calculations. To find the difference the equation $\%_{dif} = \frac{|I_{estimate} - I_{analytic}|}{I_{analytic}} \times 100$ (12), is used where $I_{analytic}$ is the result of equation (1) and $I_{estimate}$ is the result of equation (8). This tells us how different the two moments of inertia are, so they can be compared and examined and concluding why the difference may have occurred.

Materials and Method

Equipment

- Automotive crankshaft
- Cylindrical calibration mass
- Stopwatch
- Steel wire
- Ruler
- Vernier calliper

(Alam, Alluhaybi & Allhibi 2022)

Procedure

1. First the mass moment of inertia was calculated using its mass and the formula $I_{cylinder} = \frac{1}{2} \rho \pi r^4 h$ (4)
2. Rotate the calibration mass ensuring it remains stable when released
3. Average the cycles to determine the average for one period (where the mass returns to its original position)
4. Repeat steps 2 and 3 for the crankshaft
5. Calculate the crankshaft's moment of inertia by using the equations $k_{cylinder} = I_{cylinder} \left(\frac{2\pi}{T_{cylinder}} \right)^2$ (6) and $I_{estimate} = k_{cylinder} \times \frac{T_p^2}{4\pi^2}$ (8)
6. Split the crankshaft into sections and estimate the MOI using the parallel-axis theorem, approximating sections of the crank as parts of cylinders using the equations $m_i = \rho \pi r_i^2 h_i$ (9), $I_{cm,i} = \frac{1}{2} m_i r_i^2$ (10), $I_{single,i} = I_{cm,i} + m_i z_i^2$ (11) and finding the total with the equation $I_{crankshaft} = \sum_i I_i = \sum_i \left(\frac{1}{2} m_i r_i^2 + m_i z_i^2 \right)$ (2)

(Alam, Alluhaybi & Allhibi 2022)

Results and Discussion

Measured Data

Crankshaft			
Trial	Time for 5 Oscillations (s)	No. of oscillations	Period (T_p)
1	12.06	5	2.412
2	11.75	5	2.35
3	11.88	5	2.376
Mean	11.9	5	2.379

Table 1, Time trials for crankshaft

Sample Calculation

Experimental MOI

$$T_p = 2.379, \quad T_{cylinder} = 3.770, \quad r_c = 0.0755\text{m}, \quad h_c = 0.225\text{m}, \quad \rho = 7830 \text{ kg/m}^3$$

$$I_{cylinder} = \frac{1}{2} \rho \pi r_c^4 h_c = \frac{1}{2} \times 7830 \times \pi \times 0.0755^4 \times 0.225 = 8.992 \times 10^{-2} \text{ kg} \cdot \text{m}^2 \quad (4)$$

$$k_{cylinder} = I_{cylinder} \left(\frac{2\pi}{T_{cylinder}} \right)^2 = (8.992 \times 10^{-2}) \times \left(\frac{2\pi}{3.770} \right)^2 = 0.2498 \text{ N} \cdot \text{m/rad} \quad (6)$$

$$I_{estimate} = k_{cylinder} \times \frac{T_p^2}{4\pi^2} = 0.2498 \times \frac{2.379^2}{4\pi^2} = 0.03581 \text{ kg} \cdot \text{m}^2 \quad (8)$$

Parallel Axis Theorem MOI section

For section 3:

$$r_3 = 0.029\text{m}, \quad h_3 = 0.041\text{m}, \quad z_3 = 0.040\text{m}, \quad \rho = 7830\text{kg/m}^3, \quad n_3 = 4$$

$$m_3 = \rho \pi r_3^2 h_3 = 7830 \times \pi \times 0.029^2 \times 0.041 = 0.8482 \text{ kg} \quad (9)$$

$$I_{cm3} = \frac{1}{2} m_3 r_3^2 = \frac{1}{2} \times 0.8482 \times 0.029^2 = 3.5667 \times 10^{-4} \text{ kg} \cdot \text{m}^2 \quad (10)$$

$$I_{single,3} = I_{cm3} + m_3 z_3^2 = 3.5667 \times 10^{-4} + 0.8482 \times 0.040^2 = 1.7138 \times 10^{-3} \text{ kg} \cdot \text{m}^2 \quad (11)$$

Occurs 4 times in the crankshaft:

$$I_3 = I_{single,3} \times n_3 = (1.7138 \times 10^{-3}) \times 4 = 6.8552 \times 10^{-3} \text{ kg} \cdot \text{m}^2 \quad (13)$$

Sum all sections to find the analytical MOI:

$$I_{analytic} = \sum_i n_i (I_{cm,i} + m_i z_i^2) = 0.1874 \text{ kg} \cdot \text{m}^2 \quad (2)$$

Percentage Difference

$$\%dif = \frac{|I_{estimate} - I_{analytic}|}{I_{analytic}} \times 100 = \frac{|0.03581 - 0.1874|}{0.1874} \times 100 = 80.89\% \quad (12)$$

Results Summary

Quantity	Value
Experimental Moment of Inertia, $I_{estimate}$	0.03581 kg · m ²
Analytical Moment of Inertia (Parallel-Axis), $I_{analytical}$	0.1874 kg · m ²
Percentage Difference	80.89%

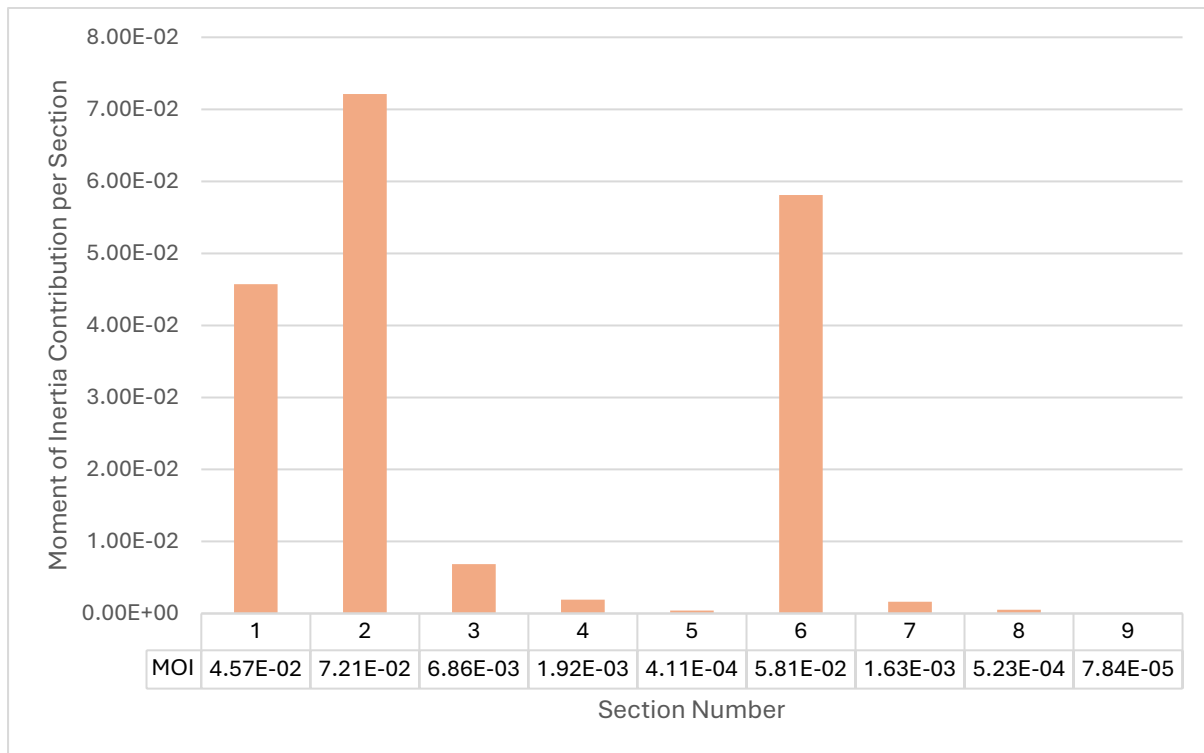


Figure 1, Contribution of each section to the total analytical moment of inertia

Discussion

From the experiment we determined the experimental (estimate) mass moment of inertia of the crankshaft to be 0.03581 kg · m² and the analytical found using the parallel-axis theorem to be 0.1874 kg · m². The difference of these is 80.89%, using the analytic value as the denominator. The analytic value was used as the reference which is 5.23× greater than the experimental value. The experimental value is more likely to be of merit in this experiment as it measures the actual mass distribution while the moment of inertia derived from the parallel-axis theorem relies on geometry. This may cause a greater amount of error in the equations.

Sections 1,2 and 6 of the crankshaft, making up 0.1760 kg · m² of the total analytic moment of inertia (0.1874 kg · m²) sits further out from the midpoint. Most of the mass being of a greater distance from the centre causes the moment of inertia to balloon. The experimental moment of inertia has a radius of 75.5 mm while the analytical moment of inertia's maximum radius is 98.5 mm, which due to the moment of inertia grows with r^4 .

The timing of the trials was done by hand with a stopwatch which may lead to greater errors as there will be a difference between the recorded time and actual trial time. This increases the period of the crankshaft relative to the cylinder because if we assume the same delay on each

mass, it makes up a greater proportion of the crankshaft's time. Furthermore, geometric measurements of the crankshaft's sections may have a small margin of error. Due to the equations' sensitivity to the size of the radius as mentioned above, this can lead to either far greater or far smaller calculations for the moment of inertia.

The moment of inertia is a vital part of engineering because it outlines the properties of force and stability needed to change the motion of an object. This matters for a crankshaft as the MOI acts to absorb torsional vibrations caused by the engine before any fatigue or failures occur. To improve our calculations for the mass moment of inertia of a crankshaft, more time trials should be undertaken with a more accurate timing method and there should be a more precise geometric breakdown of the crankshafts sections to allow for more accurate data variables to be used in calculations.

The objective of this experiment, to find the mass moment of inertia of a crankshaft using two different methods was met. Both the experiment (estimated) and analytical (parallel-axis theorem) found a valid MOI and the discrepancies between the values can be explained.

Conclusion

The mass moment of inertia was experimentally determined to be $0.03581 \text{ kg} \cdot \text{m}^2$ and analytically determined to be $0.1874 \text{ kg} \cdot \text{m}^2$, making the difference 80.89% using the analytical MOI as the subject. This vast difference in the MOI indicates that using the parallel-axis theorem under or, in this case, overestimates the I value. While there was a difference between the two values we derived, the purpose of this experiment was to measure the MOI using two different methods, which was successfully achieved. The source of the deviation was outlined in the *Discussion* as the difference in each mass' radius, exemplifying the effect the radius has in altering the MOI.

Recommendations

- Replace the manual stopwatches with a stopwatch that removes the variable of human's reaction time.
- Increase the number of oscillations in each trial, increasing the sample size and minimising fractional differences in the periods resulting from timing variables.
- Use a method to better gain the crankshaft section dimensions allowing for more accurate calculations. This could be achieved using 3D scanning or measuring displacement.

References

- Alam, F, Alluhaybi, A & Allhibi, H 2022, *Laboratory 2: Mass moment of inertia of a crankshaft*,.
- Jarrar, F 2020, *Mass moment of inertia*, viewed <<https://firasjarrar.com/wp-content/uploads/2020/05/mass-moment-of-inertia.pdf>>.
- Yamagata, H 2005, *Science and technology of materials in automotive engines: 8 - The Crankshaft*, WOODHEAD PUBLISHING.

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Appendix

Appendix A – Raw Data & Full Calculations

Cylinder				
Trial	Time for 5 Oscillations (s)	No. of oscillations	Period ($T_{cylinder}$)	
1	20.6	5	4.120	
2	18.38	5	3.676	
3	17.57	5	3.514	
Mean	18.85	5	3.770	

Table 2, Time trials for cylinder

Crankshaft in m									
Crankshaft section	Height (h) m	Diameter (d) m	radius (r) m	Offse t m	Area m^2	Volume m^3	# (n)	Total volume m^3	
	1	0.021	0.194	0.097	0	0.03	0.00	2	0.001
	2	0.018	0.19	0.095	0	0.03	0.00	4	0.002
	3	0.041	0.058	0.029	0.04	0.00	0.00	4	0.000
	4	0.039	0.068	0.034	0	0.00	0.00	3	0.000
	5	0.025	0.068	0.034	0	0.00	0.00	1	0.000
	6	0.029	0.19	0.095	0	0.03	0.00	2	0.002
	7	0.024	0.097	0.0485	0	0.01	0.00	1	0.000
	8	0.03	0.069	0.0345	0	0.00	0.00	1	0.000
	9	0.068	0.035	0.0175	0	0.00	0.00	1	0.000
						0.0026			
Total	0.295	0.969	0.4845	0.04	0.108	5	19	0.006231	

Table 3, Part A. Data and calculations for analytical moment of inertia of the crankshaft

mass				
(kg)	$I_{cm,i}$ (kg·m ²)	$m \cdot d^2$ (kg·m ²)	$I_{single,i}$ (kg·m ²)	I_i (kg·m ²)
4.860	2.287E-02	0.000E+00	2.287E-02	4.573E-02
3.996	1.803E-02	0.000E+00	1.803E-02	7.213E-02
0.848	3.567E-04	1.357E-03	1.714E-03	6.855E-03
1.109	6.410E-04	0.000E+00	6.410E-04	1.923E-03
0.711	4.109E-04	0.000E+00	4.109E-04	4.109E-04
6.438	2.905E-02	0.000E+00	2.905E-02	5.810E-02
1.389	1.633E-03	0.000E+00	1.633E-03	1.633E-03
0.878	5.227E-04	0.000E+00	5.227E-04	5.227E-04
0.512	7.844E-05	0.000E+00	7.844E-05	7.844E-05
20.742	7.359E-02	1.357E-03	7.495E-02	0.1874

Table 4, Part B. Data and calculations for analytical moment of inertia of the crankshaft

Cylinder in m							
Cylinder	Height (h) m	Diameter (d) m	Radius (r) m	Area m^2	Volume m^3	$I_{cylinder}$ (kg·m ²)	$k_{cylinder}$ (N·m/rad)
1	0.225	0.151	0.0755	0.017 908	0.004029	0.08992	0.2498

Table 5, Data and calculations for cylinder

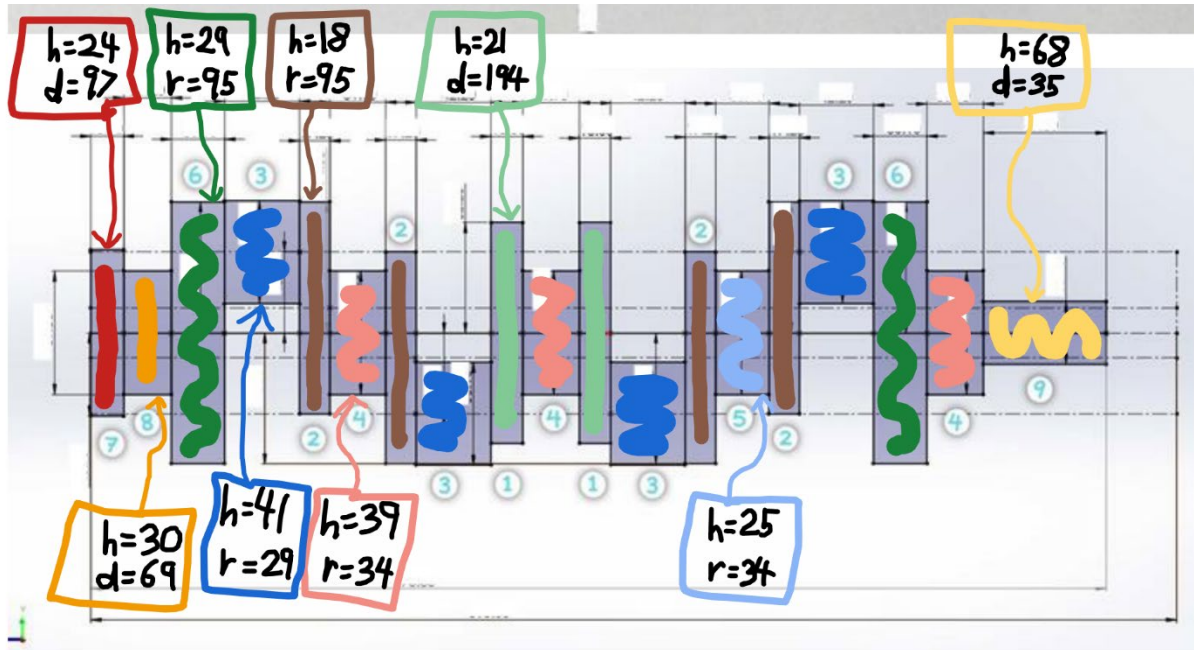


Figure 2, Dimensions of each section in the crankshaft

Appendix B – Equations

$$I = \int r^2 dm \quad (1)$$

$$I = \sum_i n_i (I_{cm,i} + m_i z_i^2) = \sum_i I_i = \sum_i (\frac{1}{2} m_i r_i^2 + m_i z_i^2) \quad (2)$$

$$I = \frac{1}{2} m r^2 \quad (3)$$

$$I_{cylinder} = \frac{1}{2} \rho \pi r_c^4 h_c \quad (4)$$

$$f_{cylinder} = \frac{1}{2\pi} \sqrt{\frac{k_{cylinder}}{I_{cylinder}}} \quad (5)$$

$$k_{cylinder} = I_{cylinder} \left(\frac{2\pi}{T_{cylinder}} \right)^2 \quad (6)$$

$$I_{crankshaft} = \frac{k_{cylinder}}{(2\pi \times f_{crankshaft})^2} \quad (7)$$

$$I_{estimate} = k_{cylinder} \times \frac{T_p^2}{4\pi^2} \quad (8)$$

$$m_i = \rho \pi r_i^2 h_i \quad (9)$$

$$I_{cmi} = \frac{1}{2} m_i r_i^2 \tag{10}$$

$$I_{single,i} = I_{cmi} + m_i z_i^2 \tag{11}$$

$$\%_{dif} = \frac{|I_{estimate} - I_{analytic}|}{I_{analytic}} \times 100 \tag{12}$$

$$I_i = I_{single,i} \times n_i \tag{13}$$